INSA INSTITUT NATIONAL DES SCIENCES APPLIQUÉES LYON

Web3 Prediction Market

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28.01.2026

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Introduction & Motivation

What are prediction markets?

Prediction markets are platforms that allow you to trade shares on specific outcomes. You place a money-backed guess to predict the outcome of events across sports, politics, economics, pop culture and more.

Source 1 - MetaMask

Exploring Decentralized Prediction Markets: Accuracy, Skill, and Bias on Polymarket

40 Pages • Posted: 13 Dec 2025 • Last revised: 23 Dec 2025

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Date Written: December 12, 2025

Abstract

Prediction markets aggregate dispersed information into probabilistic forecasts that can guide decision-making and risk management. The recent emergence of decentralized, blockchain-based platforms thus gives practitioners access to real-time probabilities for a wide range of events and, at the same time, generates unprecedented amounts of real-world trading data that are highly valuable for economic research. In this

Source 2 - Reichenbach, F. & Walther, M. (2025). Exploring Decentralized Prediction Markets: Accuracy, Skill, and Bias on Polymarket.

POLITICS • DONALD TRUMP

The French 'whale' who made massive pro-Trump bets is only doing it for the money and could win $80 million, report says

By Marco Quiroz-Gutierrez
Reporter

Add us on

November 2, 2024, 4:03 AM ET

Fig. 1 - News headline on a frenchman winning 80 million USD from prediction

Why Blockchain?

	Centralized Prediction Markets	Decentralized Prediction Markets
Trust model	Requires trust in a single authority	Trustless, based on smart contracts
Control of funds	Operator controls user funds	Smart contracts control funds
Outcome resolution	Decided by the operator	Blockchain protocols, Oracles (e.g. Chainlink)
Payout process	Controlled and executed by operator	Automatic and verifiable execution
Risk of manipulation	Yes	No
Censorship resistance	Low	High
Verifiability	Users cannot independently verify execution	Anyone can verify outcomes and payouts

Why Blockchain?

Blockchain-Based Approach

- Smart contracts enforce rules autonomously and immutably
- Trustlessness
- Transparency and verifiability
- Binding commitments
- Guaranteed outcomes

Fig. 6 - Blockchain Explorer Visualizer

Token & Betting Model

How the Market Works

- ERC-20 token used for all bets and payouts
- Yes / No outcome markets
- Users buy “shares” of an outcome
- Winnings distributed proportionally to shares

Implied odds

- Derived from share distribution
- Reflect current market sentiment (not AMM pricing)

System Architecture

Frontend

- Node JS + CSS
- MetaMask for transaction signing and validation

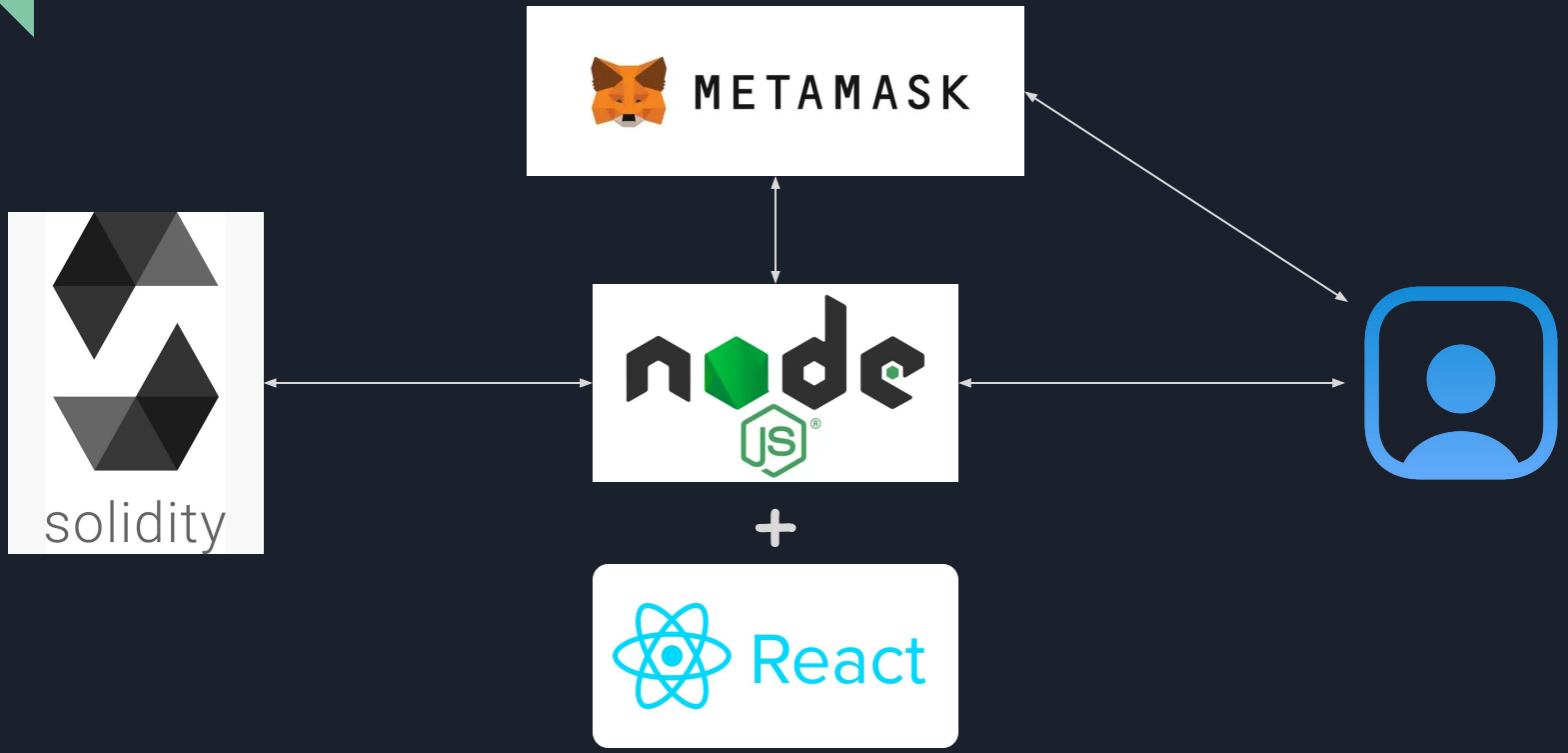
Smart Contracts

- Solidity contracts deployed via Truffle
- Enforce betting, resolution, and payouts

Local Blockchain

- Ganache for development and testing

System Architecture



Oracle & Market Resolution

What are oracles?

In prediction markets, oracles are systems that bring real-world information to the blockchain. They act as a bridge between off-chain reality and on-chain settlement.

Source 3 - MetaMask

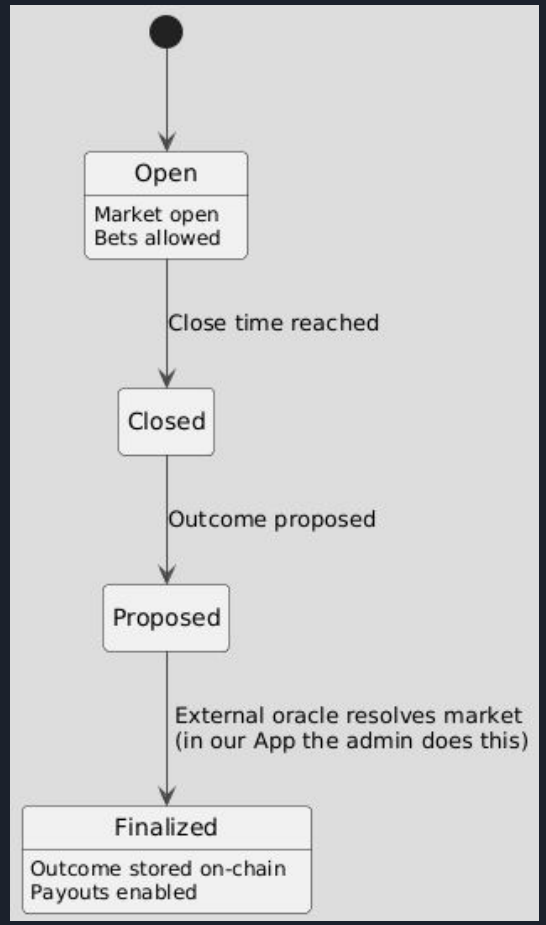


Fig. 2 - Sequence of Events Diagram

Smart Contract Design

Goals : Initialize, manage and automate the entire lifecycle of a market.

lifecycle of a market (stage) :

Active	Pending	Resolved
market is open, users can place bets	**bettings are finished, waiting for Oracle's resolution**	**outcome is finalized, winners can claim their gains**

Structures :

- **Market :** title, two possible options, end time, total liquidity pools, the current stage, creator's address.

  pools represent the total amount of Ether deposited by all the bettors on each of the options

- **Bet :** amount staked, choice

Smart Contract Design

Storage :

- store the public address of the Oracle
- array containing every market created
- mapping between the market and all the bets made on this market

The logic of the market is handle by 4 functions :

- **createMarket**
- **placeBet**
- **resolveMarket**
- **claimGain**

Smart Contract Design

1. createMarket :

- initialize market
- end time = blockchain timestamp + duration
- empty pools
- market set to **Active** stage

2. placeBet :

- payable function => allowing the contract to receive Ether
- each user can only bet on one option
- updates the global pool

3. resolveMarket :

- handles transition from **Pending** to **Resolved**
- an hash (keccak256) is formed using contract's own address, the market ID, and the outcome.
- a signature is created using this hash with the Oracle's private key
- recover function to extract the address from the signature
- if the address address matches the stored Oracle's address, the outcome is authenticated

4. claimGain :

- rewards are calculated :

$$Reward = Stake + \frac{Stake \times Losing\ Pool}{Winning\ Pool}$$

- 100% of the losing stakes are redistributed to winners proportionally to their risk
- contract resets the user's stake to zero **before** transferring any funds
- Ether is sent directly to the user's wallet

Demonstration of our Web3 Prediction Market

Trust Assumptions & Limitations

Trust Assumptions

- Oracle relies on honest participation
- Dispute resolution depends on voters

Limitations

- Only bets on one outcome are possible
- Shares can not be traded or sold before resolution
- No external data feeds

Fig. 5 - Disagreement on the outcome of the Market “Will Zelensky wear a suit?”

Future Improvements

External oracle integration

- Use of decentralized data feeds such as Chainlink and Universal Market Access (UMA)
- Reduced reliance on admin or local market resolution

Alternative betting and distribution models

- Tradable outcome positions (e.g. with ERC-1155)
- Dynamic pricing with continuous liquidity (e.g. AMMs or order book modell)
    a. Market-driven odds instead of static shares

Fig. 4- Logo of Chainlink Oracle

Fig. 3 - Logo of Universal Market Access Oracle

References

Figures:

Fig. 1 - News headline on a frenchman winning 80 million USD from prediction
https://fortune.com/2024/11/02/french-whale-polymarket-30-million-donald-trump-election-bet-kamala-harris/

Fig. 2 - Diagram showing sequence of events
created using: https://www.planttext.com/

Fig. 3 - Logo of UMA project
https://coinmarketcap.com/fr/currencies/uma/

Fig. 4 - Logo of Chainlink project
https://github.com/smartcontractkit/chainlink

Fig. 5 - Disagreement on the outcome of the Market: Will Zelensky wear a suit?
https://www.forbes.com/sites/boazsobrado/2025/07/07/the-president-wears-no-suit-polymarkets-160-million-problem/

Fig. 6 - Blockchain Explorer visualizer
https://skillfloor.com/blog/what-is-a-blockchain-explorer-and-how-to-use-it

Literature:

Source 1 - Definition of prediction market from MetaMask
https://support.metamask.io/de/manage-crypto/trade/predict/what-are-prediction-markets/

Source 2 - Reichenbach, F. & Walther, M. (2025). Exploring Decentralized Prediction Markets: Accuracy, Skill, and Bias on Polymarket.
https://dx.doi.org/10.2139/ssrn.5910522

Source 3 - Controversial Market Outcome
https://www.wired.com/story/volodymyr-zelensky-suit-polymarket-rebellion/

Source 4 - Definition Oracle from MetaMask
https://support.metamask.io/de/manage-crypto/trade/predict/what-are-prediction-markets/

References

Bonus:

DappCon Berlin - DappCon is a nonprofit global developer conference focusing on decentralized applications, tooling, and foundational infrastructure on Ethereum
https://www.youtube.com/@dappconberlin/videos